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Co-occurrence patterns based fruit quality detection for hierarchical fruit image annotation

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ABSTRACT

Automatic image annotation is a method of assigning caption to images that provide some convenient way to index, retrieve and handle a large amount of data objects. It focuses on recent agricultural automation applications; it finds potential in classification along with contextual labeling of the involved objects or detailing based on its statistical properties on fruit categories. However, producing hierarchical labels provide details of a particular fruit subcategory. This paper proposes fruit annotation in a broad sense along with its hierarchical features that can be narrowed down to inherit, further achieving fruit classification into binary or multiple classes indicating subcategories of that fruit. The fruit objects within images are measured to its actual size in the required units. The classification is also used for identifying true color, texture, size, deep features and shape based on the ratio of major to minor axis helpful for fruit gradations. The co-occurrence patterns are obtained based on the visual features of the selected fruit. This is useful for finding the fruit quality categories and combined properties that are used to form the co-occurrence patterns. These patterns are further used by the classifier for fruit annotation. The evaluation of the performance is carried out using the F1 score, accuracy, precision, recall and G-measure. The results show that the co-occurrence pattern with SVM provides an overall accuracy of 97.3% and 97.2% for grape and mango fruit subcategories. The comparative results are obtained to cross-check with the subjective evaluation of gradation validated by local farmers.
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1. Introduction

Computer vision (CV) plays a vital role in industrial automation. CV based agricultural automation applications such as fruit image annotation is difficult due to color, texture, shape, and size similarity between various fruit species and also a high variability in a class. Therefore, manual fruit annotation is a time consuming and tedious task. Here, automatic fruit image annotation plays an important role in academic research and industrial applications. Automatic fruit image annotation is of assigning a caption to fruit images which provides some convenient way to index, retrieve and handle a large number of fruit categories. Several useful application systems can be built such as supermarket price detection (Deliens et al., 2016), dietary guidance (Di Cagno et al., 2017), and fruit packaging and transportation (Getahun et al., 2017) to support industry. The supermarket is one of the applications of fruit annotation where the cashier needs to recognize a fruit species category for determining the correct price stored in the database. Nowadays, a barcode is assigned to each fruit species based on their variety. Customers prefer to choose fruits on their own. Annotation-based applications can detect the fruit category purchased by the customer and retrieve its correct price from the database of the supermarket. Again, by using automatic fruit annotation, human expert's subjectiveness can further be reduced. Currently, fruit classifications are of two types; one is to classify various fruit categories and the other is to identify a particular type of fruit (Zhang et al., 2019).

A significant interest in the development of computer-based automatic fruit annotation is increasing in the research community. Recently, deep learning was successfully applied to the agriculture domain (Sa et al., 2016; Kamilaris and Prenafeta-Boldú, 2018).

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Especially, a convolutional neural network (CNN) has shown superior performance than existing techniques used in CV problems such as image annotation and classification tasks including fruit recognition problems (Mureşan and Oltean, 2018; Mahendran et al., 2012).

The methodologies mentioned in the literature survey only think about the fruit categories, however, producing hierarchical labeling provides details of a particular fruit subcategory. The classification of subcategories contains describing mutually exclusive images as per their features and categorizing an unseen fruit image into its subcategories by using these multiple features. The extracted features such as shape, color, size and texture features could be feeble as an individual feature, however, if combined together would become robust and discriminative features (Yuan et al., 2011). The features that are combined based on co-occurrence patterns (CP) are effective for fruit quality classification. This CP can increase the strength of classifying the quality detection of fruit. The obtained features are given to the classifier which further generates a concept signature. The concept signature stores the low level and semantic content of an image. Based on the similarity of the semantic concept, the image is matched and retrieved. Therefore, the concept signature vector reduces the memory cost required for stowing a large number of images and thus further reduces the computational cost. For experimentation purpose, the CNN model is selected based on architecture and popularity among the research community. The extracted features from the CNN model and CP that are formed are used for boosting the result of classification. From the literature, it is observed that the CP concept is not applied for fruit annotation with quality grading, which is attempted for the first time in this study.

Western Maharashtra has the largest economy of India mainly due to the availability of fruits such as grapes and mangoes. The ideal geographical location for the grapes, especially in Sangli district and mangoes in the Konkan region such as the Ratnagiri, Devgad districts, etc. Therefore, these particular fruits are selected for research purposes. Thus, there is deep to pay attention to the above fruit categories regarding agricultural automation application focuses on the mentioned fruit categories. The quality detection of the above fruits not only helps to increase productivity, but it will also segregate the identification of the best quality fruits in the context of the specific farmer. This will lead to an increase in the economical benefit of the actual fruit farmers who are associated with these fruit industries.

The experimental results show that the proposed CP for fruit quality detection and fruit annotation is a non-destructive method and effective for the categorization of subcategories of the fruit. The system is reliable, accurate to verify the type of fruit viz. table grapes and varieties of mangoes. In the fruit industry, mostly, the quality grading verification with categorization is still accomplished manually. High-grade products lay the foundation for achievement in today's competitive market. Therefore, such an approach can be useful for in-line sorting fruit-based systems. The main contributions of this work are condensed as follows.

1. In this paper, measures are extracted for identifying the true color, size and shape based on the ratio of the major axis to the minor axis of a fruit image, color histogram, color moment, texture feature by combining discrete wavelet transform (DWT) and Gabor filters with a band of 51–75% (Nemade and Sonavane, 2020). The deep learning model, ResNet50 (He et al., 2016) is used for the automatic extraction of features. The principal component analysis (PCA) is employed for dimensionality reduction of deep features.
2. The co-occurrence patterns are obtained based on the features of the selected fruit and are used for the detection of the fruit quality and its subcategory. The classification is carried out using softmax and SVM classifiers.
3. The results of the experiment can effectively improve the quality detection and the fine-grained fruit annotation of the classifier using the softmax and SVM classifier.

The rest of the paper is organized as follows: Section 2 provides related work; Section 3 describes the architecture, fruit image quality classification, CNN model, classification methods and complexity analysis. Section 4 contains the dataset and results and discussion. The conclusion is given in Section 5.

2. Related work

For the classification and annotation of fruits, there are various studies in the literature based on the image visual features such as color, texture, shape and size. Seng and Mirisaee (2009) have devised a method that combines features like RGB color average value, round shape and size (by area and perimeter) features to improve the accuracy of fruit recognition. A variety of fruits can be recognized based on color histograms made up of selected intensity values of RGB colors (Rachmawati et al., 2015). From the RGB histogram, relevant features are selected using the chi-square method. K-NN classifier is used to classify these features. Lu et al. (2016) combined the color, shape and texture features of the fruit and the applied PCA for feature reduction. This reduced feature vector was given to the HPA-SLFN classifier which achieved 89.5% accuracy. However, green grapes and passion fruits are not identified well by this method. Based on some features of dates, Reem et al. (2018) have classified dates using the SVM classifier. Effective classification methods based on the color, shape and scale-invariant transform features of apple, strawberry and orange fruits are extracted and classified using random forest (Zawbaa et al., 2014). Types of fruits that are almost similar in color and shape are not considered for testing. Another approach suggested by Naskar and Bhattacharya (2015) for a fruit recognition method that focuses on color, texture and shape features, wherein fruit color is identified by mean hue values, texture using a log Gabor filter and shape using area and perimeter of a fruit. A neural network classifier is used to classify these features. Shape-based fruit recognition and classification methods are invariant to the growth stage of fruit, varying the distance and surface appearance of fruit images are discussed in (Jana et al., 2017). Using region-based and boundary-based secondary feature and object contour model, Roomi et al. (2012) have classified three sub-categories of mango fruit using Bayes classifier. For fruit classification application, Wang et al. (2015) have employed principal component analysis, wavelet entropy (WE), feedforward neural network and biogeography-based optimization.

Recently, Castro et al. (2019), have used the PCA method to combine HSV, RGB and L*a*b* color space with the combinations of K-nearest neighbors, decision trees, artificial neural network and support vector machine for classification of cape gooseberry fruits. The result showed a performance increase due to seven principal components. For the detection of fruit peel defects, Woźniak and Połap (2018) have developed a hybrid model where pre-detection is done using adaptive neural network method and peel defect recognition is performed using heuristic defect detection method respectively. In the first stage, every individual segment of the fruit image is trained and classified with possible defects. In the second stage, the defective segments are given to the heuristic model which seeks for the pixel features that are defined as defects in the fitness condition. Capizzi et al. (2015) have presented the RBPNN model for finding the class category of an orange fruit based on the defective surface of an orange. The color and texture features are computed and classified using the RBPNN model for performing the categorization task. For grape fruit size detection

and grading, Qiaohua et al. (2017) have designed the system using an improved version of the least medium square, which achieved good accuracy. The elliptical basic neural network classification technique is applied to the combination of co-occurrence matrix-based texture features and SVD for recognition of defects in organic solar cells (Sciuto et al., 2019).

The literature mentioned above has used handcrafted features and simple structured classifiers that are not able to map the complex features with good accuracy. For these reasons, nowadays the research community is focusing more on bringing CNN into play. It is very well-known for image annotation and classification tasks. It minimizes the error as it learns the features automatically and extracts it globally with contextual details. For the object recognition task, it has achieved outstanding performance on the ImageNet dataset (Krizhevsky et al., 2012). It is applied for image classification to classify the given image as a food image or not a food image and also used for recognizing the category of food using the GoogleNet pre-trained model (Singla et al., 2016). Hou et al. (2016) have designed an algorithm for fruit recognition using CNN. In this paper, the author has used a selective search method for the extraction of the image region, then the region is selected using the mean entropy of fruit image and such regions are given as an input to the CNN learning. Zhang et al. (2019) have revealed that the overall accuracy of image-based fruit classification can be increased with data augmentation techniques. With the custom-designed of 13 layers CNN, their model has achieved 95% accuracy. All these methods worked on identifying the fruit images and their quality, but very few literatures have worked on hierarchical fruit categorization.

3. Methodology

This section briefs about the overall architecture of the fruit image annotation (Fig. 1). First, images are organized from the collected image samples (Fig. 2) and then image preprocessing techniques such as noise elimination, color segmentation and shadow removal are applied to enhance the input images. Further, the actual size of the fruit image in millimeters (mm) is calculated (Fig. 4).

The shape of the fruit is evaluated based on the major and minor axes. The texture features are obtained using the hybrid DWT and Gabor filter. Subsequently, the co-occurrence pattern is created using the color, shape, texture and actual size and deep features of the fruit image. The obtained CP is used for the classification of fruit quality. Two different classifier models such as softmax and SVM are used for classification purposes. Statistical analysis is carried out using the one-way ANOVA technique.

3.1. Image pre-processing

Image pre-processing is an essential step towards improving image quality, such as reducing noise in the image and enhancing the brightness of the image, thereby increasing the accuracy of the feature extraction (Mercol, 2010).

3.1.1. Color segmentation and shadow removal

RGB color space describes the combination of red, green, and blue colors present in an image. It is more sensitive to the variance of illumination. Therefore, it is transformed into HSV color-space with the help of MATLAB toolbox inbuilt function 'rgb2hsv' as HSV color space is less prone to the variance of illumination. In HSV space, foreground objects (that are fruits) can be separated from background objects by applying an appropriate threshold on hue and saturation. Shadow effects are removed by setting a threshold as saturation value to 40 (%) and 60 (%) for grapes and mangoes respectively. Then, the image is converted into a binary image where the foreground object is white while the background is black using a suitable threshold as per Eq. (1).

$$h(\mathrm{m},\mathrm{n}) = \begin{matrix} 1 & if\ g(m,n) \geq T \\ 0 & if\ g(m,n) < T \end{matrix} \tag{1}$$

where,

h = Segmented binary image

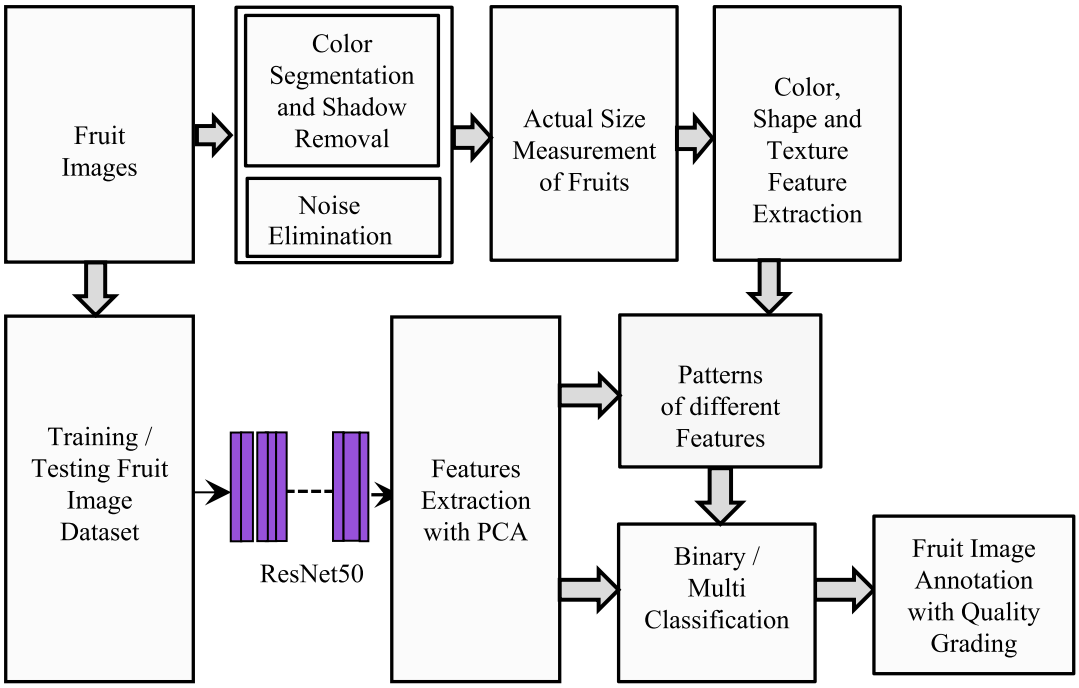


Fig. 1. The proposed model workflow for fruit image annotation.

Fig. 2. The sample of hierarchical fruit images.

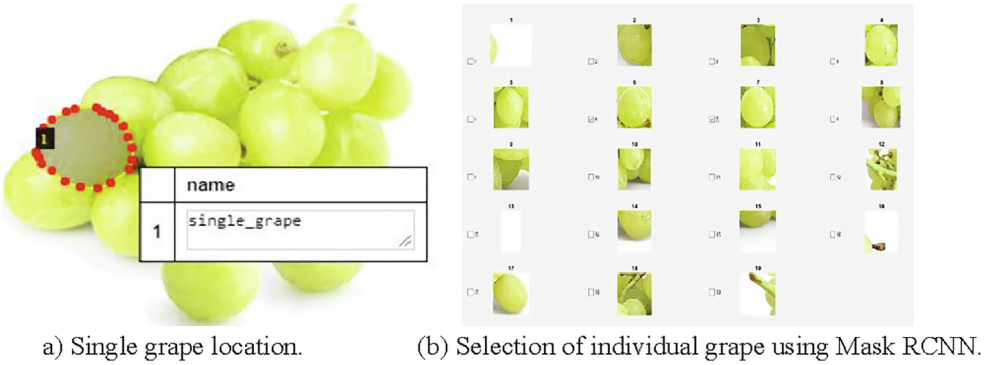


Fig. 3. (a, b) Grape Selection.

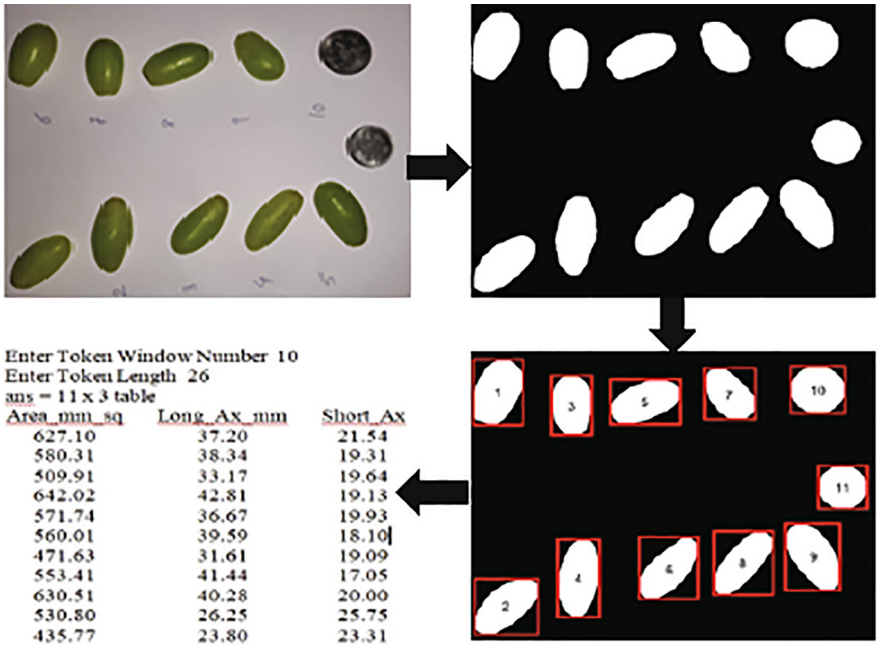


Fig. 4. The process involved for actual size measurements of fruits.

(m,n) = Pixel position
T = Threshold

3.1.2. Noise reduction

The noise of an image is removed with the help of the morphological opening operation. It is carried out to remove the narrow bridges and minor extensions that are present in the object. It also smoothes the boundary of the object. Morphological opening operation is a combination of erosion (⊖) followed by a dilation (⊕) operation with the same structuring element. The opening operation is given in Eq. (2),

$$X^{\circ}Y = (X \ominus Y) \oplus Y \tag{2}$$

where

X is an image
Y is the structuring element with disk two (a radius of two pixels).

Subsequently, the area smaller than 100 pixels is removed from the obtained binary image. After applying these steps, the image noise is eradicated.

3.2. Actual size measurement of fruits

In this study, the table grape and mango fruit subcategories are identified. For proper identification of fruit quality and category, it is necessary to find the features of each fruit berry. Generally, table grapes fruits come up in a bunch, so it is very difficult to separate berries from these fruits using traditional image segmentation techniques such as watershed transform, edge-based segmentation and so on. Therefore, after experimentation on various image segmentation techniques, it is found that the Mask R-CNN method works well for these types of problems (He et al., 2017). Hence, it is employed in this study. It detects individual instances in an image. The preprocessed image, shown in Fig. 3(a), is given as an input to the Mask RCNN. From the result of the Mask RCNN, shown in Fig. 3(b), the individual berry is picked up for finding its actual size in mm, for instance, the number 17 is picked up and its minor axis length and major axis length, the centroid and area are evaluated using the region prop function. All these computed parameters are in terms of pixels. The bounding box is drawn over the objects and the number is assigned to each bounding box. The diameters of the objects inside the bounding box are computed in vertical and horizontal directions. The longest diameter is called as major-axis while the shortest diameter is denoted as a minor axis. The aspect ratio of the major axis length and minor axis length is measured and the area is calculated by counting the number of pixels inside the boundary.

For calculation of the actual size of a fruit, the relative grading method is used, where the reference coin is utilized in the fruit image. The coin diameter is obtained with the help of a vernier caliper. The measured coin diameter is given as an input to the system and then a bounding box appears on the fruit object that is represented by a red rectangle then the user selects a token area (for example 10 specified in Fig. 4) and provides the diameter of the coin as an input to the system. Based on the diameter of the coin, the system automatically generates the area, major-axis and minor-axis of fruits.

3.3. Color and shape feature extraction

The color feature is widely used due to its invariance property with respect to image rotation, translation and scaling. This paper uses color histogram and color moment features. The RGB color histogram provides information about the distribution of the RGB color in an image. Based on the experimentation, it is found that the RGB histogram provides better differentiating colors than other color spaces. The histogram shows the occurrences of each pixel in an image. Many indications regarding the image can be drawn from the shape of the histogram. For experimentation, the sample is picked up randomly from fruit images. Each color space of the sample is then separated. Thereafter, the histogram of each color space is calculated. Table 1 shows the histogram of the sample that is taken from the fruit image and the distribution of each color component of the sample.

The color histogram provides the global distribution of color, so it is not able to find the spatial relationship among colors. To overcome this, color moments are computed to differentiate sample images based on their color features. Therefore, the color moments such as color mean and standard deviation are calculated using Eqs. (3) and (4).

$$mean_i = \frac{1}{N}\sum_{j=1}^{N} X_{i,j} \tag{3}$$

$$\text{standard deviation}_i = \sqrt{\left(\frac{1}{N}\sum_{j=1}^{N}\left(X_{i,j} - mean_i\right)^2\right)} \tag{4}$$

where, X is an image and (i,j) are image pixel positions.

The color moment provides a compact representation of a descriptor. The samples are taken from each grape category and subsequently, their standard deviations (SD) are calculated. Afterward, the mean value of the standard deviation is computed and compared with the SD of each sample taken from the category that gives the absolute difference. To find the color uniformity of the test image, the absolute difference must be as small as possible.

In this paper, the centroid and aspect ratio of major axis length to minor axis length features are utilized to describe the shape of a fruit. The centroid feature is the center position of an object which is fixed for any shape. To determine whether the fruit object is round or oval, the ratio of the major axis and the minor axis is measured. If the ratio is greater than one, then the shape is oval, otherwise, it is round.

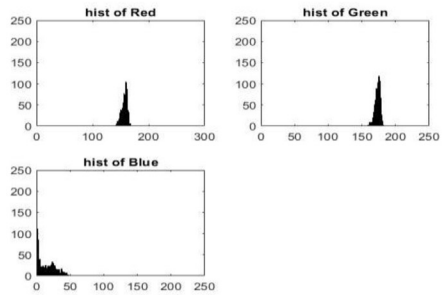
3.4. Stem color extraction

For the RGB stem sample, a histogram is plotted for each r, g, b color. Afterward, from the histogram of red and green color, the count and bin locations are computed. Histogram bins are the range of values. If the green bin locations are between the threshold values of BA and BB (for instance, BA = 150 and BB = 200) and the count is greater than one, then the stem is categorized as a green stem. Otherwise, if green bin locations are greater than the threshold value BC = 50 and red bin locations are less than the threshold value BD = 100 and the count is greater than one, then the stem is categorized as the brown stem.

3.5. Texture feature extraction

Texture features contain valuable information therefore these are suitable for classification and fruit quality assessment (Kondo et al., 2000; Arivazhagan et al., 2010). It separates various patterns in fruit images with the extraction of pixel intensity values. These are helpful for predicting surface appearance in terms of smoothness or roughness. The texture is well-match to ripeness and sugar content (fruit internal quality). There are four types of texture features available such as statistical, model-based, structural and transform-based textures (Bhargava and Bansal, 2018). The statistical methods require minimum computational cost but have constraints toward spatial relations over small neighborhoods on a

Table 1
RGB histogram and color moment of the image sample.

Image	RGB histogram	Standard deviation
		13.2557

single scale only. Therefore, mostly a transform-based feature extraction technique is used.

3.5.1. DWT

DWT is used for image processing as it catches spatial as well as frequency domain information. For effective analysis, DWT takes the image size in dyadic, that is, 2 k form. Therefore, every image is resized to 256*256 with k (eight) level decomposition of an image.

In this study, DWT (db4) is selected for decomposition due to its well-known smoothing and orthogonality properties for texture feature extraction (Pujari et al., 2013). DWT decomposes the image into four different subbands of low pass (approximation coefficient) and high pass (detail coefficients), namely LL1 (approximation coefficient) and three detail coefficients as LH1 (horizontal), HL1 (vertical) and HH1 (diagonal) information of an image. Utmost information is present in the LL1 sub-band but it does not provide any information regarding image singularity, so this band is utilized for the next level of decomposition just like the original image. The approximation coefficient of the next round, LL2 sub-band, is extracted for the next level decomposition.

The output of each detail coefficient after the fourth level is given to the Gabor filter where the frequency ranges from 51 to 75% is utilized for texture feature extraction.

3.5.2. Gabor filter

Gabor filter has revealed improved multi-resolution abilities because of multiple scales and orientations. Moreover, it has an ability of the optimum localization of features in the spatial as well as a frequency domain. It is a very popular technique for the extraction of texture features from images. It can obtain effectual texture features as compared to other transforms (Alimohamadi and Ahmadyfard, 2009). The 2-D Gabor function is computed via Gaussian envelope with the sinusoid frequency w and standard deviations σx and σy that are specified below:

$$g(x,y) = \frac{1}{2\pi\sigma_x\sigma_y} e^{\left(-\frac{1}{2}\left(\frac{x^2}{\sigma_x{}^2}+\frac{y^2}{\sigma_y{}^2}\right)+2\pi jw_x\right)} \tag{5}$$

In the experimentation, lower cut-off frequency (fl) = 0.51 and higher cut-off frequency (fh) = 0.75 are chosen. The DWT based processed images (detailed sub-images) are individually convolved with the Gabor kernel coefficients as stated in Eq. (6).

$$G_{ab}(x,y) = \sum\nolimits_s \sum\nolimits_t I(x-a, y-b) g^*_{ab}(m,n) \tag{6}$$

In the equation, I(x,y) is the HL, LH, HH sub-band images of DWT, the complex conjugate g^*_{ab} of Gabor function, (m, n) denotes kernel size and G_{ab} is the convolution output.

The statistical measures such as energy (E_{ab}), mean (μ_{ab}), standard deviation (σ_{ab}) and skew (α_{ab}) and kurtosis (γ_{ab}) are computed using Eqs. (7)–(11) for characterizing the texture features of images.

$$E_{ab} = \sum\nolimits_x \sum\nolimits_y |G_{ab}(x,y)| \tag{7}$$

$$\mu_{ab} = \frac{E_{ab}}{P * Q} \tag{8}$$

$$\sigma_{ab} = \frac{\sqrt{\sum_x \sum_y (|G_{ab}(x,y)| - \mu_{ab})^2}}{P * Q} \tag{9}$$

$$\alpha_{ab} = \frac{\left(\sum_x \sum_y (|G_{ab}(x,y)| - \mu_{ab})^3\right)^{1/3}}{P * Q} \tag{10}$$

$$\gamma_{ab} = \frac{\left(\sum_x \sum_y (|G_{ab}(x,y)| - \mu_{ab})^4\right)^{1/4}}{P * Q} \tag{11}$$

$$F_v = (\mu_{ab}\sigma_{ab}\alpha_{ab}\gamma_{ab}) \tag{12}$$

These features are combined in the feature vector F_v, and therefore total twenty-four features are obtained in the feature vector F_v. Since the extracted features from images are represented through statistical parameters which can be considered as multi-dimensional points, therefore the distances between these corresponding points can be calculated. The Indian statistician, P. C. Mahalanobis has introduced the Mahalanobis distance (MD) for determining a given sample, is a member of a category, or is an outlier (Mahalanobis, 1936). Mahalanobis distance is used before classification to measure the distance between two points in multi-dimensional space as shown in Fig. 5.

Mahalanobis distance is defined as:

$$MD = \sqrt{(x1 - mn)^T S^{-1} (x1 - mn)} \tag{13}$$

In the equation, a row vector x1 contains observations, mn gives the mean of the given reference sample and S is the variable covariance-matrix. The MD is the Euclidean distance marked as per the two cluster's standard deviation. In order to apply MD, to classify the reference sample, the covariance matrix of a class is evaluated and then, MD of a class is computed. Finally, the reference sample is classified as smooth or textured depending on the minimal MD.

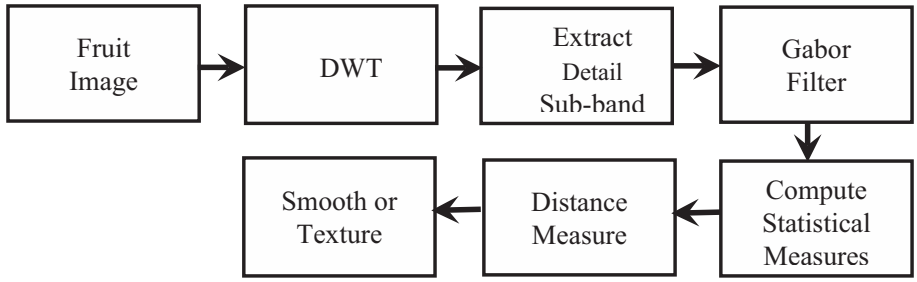


Fig. 5. DWT and Gabor methodology for texture feature extraction.

3.6. CNN

After a literature survey, it is found that deep features are well-functioning for classification problems. Therefore, in this study, the pretrained CNN model is utilized for the extraction of features. The features are taken from the global average pool layer of ResNet50 and given to PCA for dimensionality reduction.

3.6.1. ResNet50 (Residual Network)

ResNet50 is trained on the ImageNet dataset and has 50 layered residual networks. In the five stages of the convolution layer, the first layer 7 × 7 convolution matrix is applied with a stride of two; therefore, like the pooling layer, the input is downsampled by two. The next three identity blocks are followed before downsampling by two. This procedure continues for several layers. Every residual module has a stack of three (1 × 1, 3 × 3 and 1 × 1) convolution layers. The first and last 1 × 1 convolutions are employed for enhancement and dimensionality reduction. Skip connection is utilized to add the output of the previous layer to the next layer, which helps to recognize global features and minimizes the vanishing gradient problem. Average pooling is used in the last layer which produces thousands of feature maps. It extracts complex descriptors and retains the simple features of the initial layers that accelerate convergence.

3.6.2. Principal component analysis (PCA)

It is a dimensionality reduction technique that reduces the number of features, removing possible correlation, however, keeping the important information of input data (Chan et al., 2015).

3.7. Fruit image quality classification

When more than two properties of an object occur together, then that forms some pattern, that pattern is known as the co-occurrence pattern. As shown in Fig. 6, the co-occurrence patterns are formed based on feature extraction such as size, color (fruit and stem), shape and texture (smooth and texture) for the quality categorization of fruit images (see Fig. 7).

Suppose N features co-occurrence patterns are considered, then the quality category of fruit will be given as per Table 2. As observed in the table, category-A+ stands a very good category, category-A is also very good but little less quality than category-A+, likewise, further categories are quality-wise differentiated. If appearance/feature is textured one (>10%) then it takes precedence over size, color, shape, stem color and considered as rejected since it is non-confirming to quality. The detail algorithmic steps are given in Table 3.

3.8. Classification

The classification of fruit quality grade with annotation using softmax and SVM classification techniques.

3.8.1. Softmax classification

Using the softmax function, the probability of the nth class of image X_m can be calculated as:

$$P_{m,n} = \frac{exp(q_n(X_m))}{\sum_{n=1}^{C} exp(q_n(X_m))} \tag{14}$$

In the equation, $q_n(X_m)$ is the discrete probability distribution of the image X_m for the nth class concept.

To reduce the KL divergence between ground truth and prediction probabilities, the softmax loss function is given as:

$$\mathrm{fl} = -\frac{1}{N}\sum_{m=1}^{N}\sum_{n=1}^{C}\overline{p}_{m,n} log(p_{m,n}) \tag{15}$$

where, $\overline{p}_{m,n}$ is the ground truth for the class n of image X_m that is, if $\overline{p}_{m,n}$ = 1 means the label of the image is present and if$\overline{p}_{m,n}$ = 0 means the label of the image is absent and fl is the cross-entropy loss. The ground truth vectors are fixed throughout the complete training process. To reduce the loss of cross-entropy, the network is trained in such a way that the output vector q_n is near to its related one-hot vector.

3.8.2. Support vector Machine (SVM)

The SVM is introduced by Vapnik (1998) for binary classification problems. It aims to search the optimal hyperplane $f(W.X) = Y = W.X + b$ to distinguish classes with features $X \in R^m$. The label is shown by $Y \in \{-1, +1\}$. The problem is mathematically formulated as:

$$\begin{aligned} &minimize: \ \tfrac{1}{2}\|W\|^2 \\ &Subject\ to: \ W.X + b \geq +1\ for\ Y = +1 \\ &W.X + b \leq -1\ for\ Y = -1 \end{aligned} \tag{16}$$

The SVM classifier determines a hyperplane for separating classes in a multi-dimensional feature space. It searches the hyper-plane that maximizes the width between support vectors and the hyperplane. It separates the data on decision boundary and non-linear boundaries by applying some kernel tricks. Several kernel functions are used by SVM such as RBF (Radial Basis Function), polynomial, linear and sigmoid. These kernel functions allow searching for multiple hyper-plane. It can be used for multiclass problems via several methods like one-against-one, directed acyclic graph (DAG) and one–against-all.

3.9. Complexity analysis using mathematical model

The complexity of the algorithm costs $O(n)$ for computing the CP of various features and for computing the quality grade it takes $O(n)$. The deep learning network with softmax requires $O(p \times \mathrm{q})$ time, where p and q are the number of parameters (trainable) and the dataset size, respectively. If the value of p is extremely high, then the computation overhead increases. The CP feature set is obtained as per Table 3 algorithm which is given to the clas-

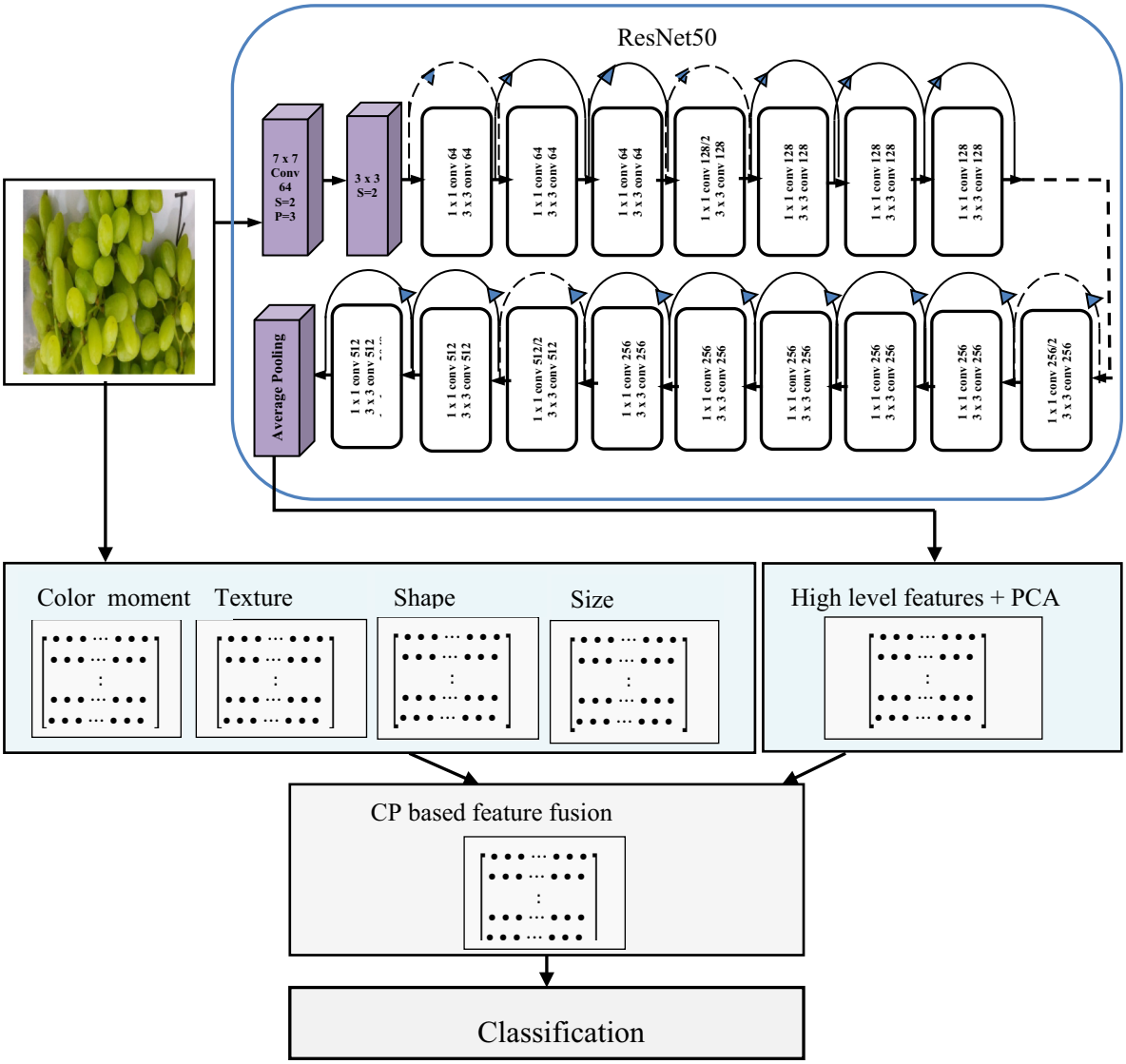


Fig. 6. Feature fusion based on CP.

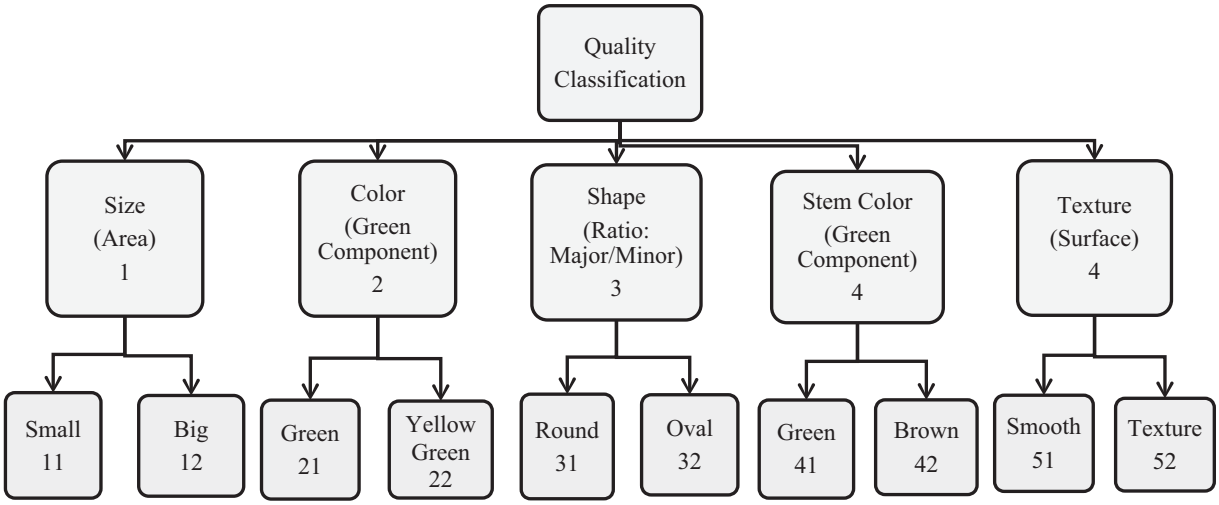


Fig. 7. Fruit image quality classification.

Table 2
Grape quality categorization based on co-occurrence pattern.

Sr. No.	Cooccurrence pattern	Description	Quality Category
1.	11-21-31-41-51	big-green-round-green_stem-smooth	A+
2.	11-21-32-41-51	big-green-oval-green_stem-smooth	
3.	11-22-31-41-51	big-yellow-round-green_stem_smooth	
4.	11-22-32-41-51	big-yellow-oval-green_stem-smooth	
5.	11-21-31-42-51	big-green-round-brown_stem-smooth	A
6.	11-21-32-42-51	big-green-oval-brown_stem-smooth	
:	:	:	:
:	:	:	:

sifier. For softmax classification, the combined feature operation is as follows:

$$CF = max(0, \sum_i^n w_i t_i + \sum_j^n w_j h_j + b) \tag{17}$$

where $\{t_1, t_2, ..., t_i, ..., t_n\}$ are CP_based features, $\{h_1, h_2, ..., h_i, ..., h_n\}$ are deep features and the bias b.

Thus, this study also utilizes a classification layer that reduces the trainable CNN parameters and increases training efficiency. During the label assignment with quality grading, it takes $O(n)$ time for all n test images. For one-vs-all SVM classification (Lingras and Butz, 2007) using a parallel environment, given a d-dimensional vector and the training dataset N which contains CP features and its corresponding label is described in the form of:

$$\{(s_x, p_x) | s_x \in R^d,\ p_x \in \{1, 2, ..., C\}\},\ x = 1, 2, ..., N \tag{18}$$

In the given Eq. (18), s_x is the d-dimensional vector, p_x represents a class label and C represents the total number of classes. The classification function for each binary SVM is given as:

$$f_n(s) = \sum_{x=1}^{N} p_x^n \alpha_x^n k(s_x, s) - b_n,\ \text{n} = 1, 2, ..., \text{C} \tag{19}$$

$$p_x^n = \begin{cases} +1 & if\ s_x \in n^{th}\ class \\ -1 & otherwise \end{cases} \tag{20}$$

$p_x^n \in \{+1, -1\}$ depends upon s_x class label. $p_x^n = +1$ if s_x has $n^{th} class$ label, otherwise $b_x^n = -1$. The kernel function, Lagrange's coefficient and bias are represented by k(), α_x^n and b_n respectively. α_x^n and b_n are acquired through training the nth individual SVM. The SVM output (Y) is given as:

$$Y_n(s) = sgn(f_n(s)) \tag{21}$$

If $f_n(s) > 0$ then $Y_n(s)$ is +1 that signifiess belongs to $n^{th} class$ else s belongs to a different class.

4. Experimental results and discussion

Experiments are conducted with a configuration of a 64-bit Intel-i7 processor having an operating speed of 2.80 GHz. MATLAB R2019a is selected for the implementation of this system. The dataset is arbitrarily partitioned into the training (60%) and testing (40%) set, which perceived better accuracy. The 10-fold cross-validation is applied to the training dataset. The metrics of evaluation used for this experimentation are precision, recall, accuracy, F1 score and G-measure for the performance of fruit image annotation (Tharwat, 2018).

The accuracy of the various models is calculated based on the percentage of correctly classified labels of the test dataset.

$$Accuracy = \frac{\sum_{p=1}^{n} TP_P}{\sum_{p=1}^{n} \sum_{q=1}^{n} N_{pq}} \tag{22}$$

where N_{pq} represents the classifier's total prediction, TP_P – True positive (the number of class p samples that are correctly classified as class p).

Precision is defined as a proportion of accurately predicted labels to the total number of labels predicted by the CNN model.

$$Precision_p = \frac{TP_P}{TP_P + FP_P} \tag{23}$$

where FP_P represents the False positive (the number of samples that are not from class p and wrongly classified as positive class p).

The recall is described as a proportion of accurately predicted labels to ground truth labels.

$$Recall_p = \frac{TP_P}{TP_P + FN_P} \tag{24}$$

where FN_P represents the False negative (the number of samples that are from class p and wrongly classified as negative class p).

F-measure is also called a $F1$ score, which is the harmonic mean of precision and recall and G-measure, square root of precision and recall, are calculated as per Eqs. (25) and (26) respectively.

$$F1\ score_p = 2 * \frac{Precision_p * Recall_p}{Precision_p + Recall_p} \tag{25}$$

$$G - measure_p = \sqrt{Precision_p * Recall_p} \tag{26}$$

4.1. Dataset

The hardware that constitutes this system is as follows:

- Camera model: One Plus6 Mobile
- Resolution: 1080 × 2280 and the mobile is located 30 cm above the sample.
  i. A sample of green grape fruits is collected from a site located in Miraj, a city in southern Maharashtra. The samples consist of 400 table grape fruits and have two subcategories (Amber and Milky-green grapes). Images are cropped for proper alignment. Grape fruit images were collected in December 2018.
  ii. The fruit dataset (which contains species of mangoes) is collected via digital camera, from village Ambrad, located in Taluka Kanakavali, District Sindhudurg and the local market of southern Maharashtra. The mango fruit dataset comprises 1300 images from nine different categories of mangoes images. Nine distinct categories of the mangoes are Badami, Baingan_phali, Dashehari, Gulab_khas, Hapus, Kesar, Langra, Rajapuri and Totapuri. Mango images were collected in May 2019. The representative sample fruit images are shown in Fig. 8.

4.2. Results and discussion

Here, the experimentation results of the work are discussed. Firstly, the performance comparison of transformation-based texture feature extraction techniques is explored. Next, inline fruit sorting is implemented using various classification methods (ResNet50, CP_SVM and CP_Softmax) and which type of classification method is better for the grape and mango fruits category. The pretrained CNN ResNet50 model is compared with the proposed

Table 3
Algorithm.

Algorithm: Formation of co-occurrence pattern algorithm for the quality category of fruit.

Input: Image dataset.
Output: Quality categorization based on co-occurrence pattern, g(.): x→{A+, A, B, C, Rejected}.

```
Measure the size of grape berry using the relative grading method.
Compute
    mean= 1/N Σ_{j=1}^{N} X_{i,j}
    standard deviation = sqrt( (1/N Σ_{j=1}^{N} (X_{i,j} − mean_i)^2) ) of grape berry fruit color
    sample and compared with the reference.
Compute ratio = (major_axis/minor_axis)
    if (ratio > 1) then
        shape ← 'oval'
    else shape ← 'round'
Set the threshold values as BA ← 150, BB ← 200, BC ← 50, BD ← 100
Compute the histogram of RGB color space
    if ((green_bin == BA) and (green_bin == BB)) and (count > 1)) then
        stem ← 'green_stem'
    else if ((green_bin >= BC) and (red_bin < BD)) then
        stem ← 'brown_stem'
Compute image texture
    Apply DWT (DB4) to decompose the gray level image into approximate
    and detailed sub-bands up to the fourth level.
    Extract HL, LH and HH fourth level sub-images (sub-bands).
    Apply Gabor filter with 51% to 75% frequency band for extraction of
    features on these sub-images individually.
    Calculate statistical parameters such as mean, standard deviation, skew
    and kurtosis using Eqs. (8)–(11) for characterizing the texture features.
    Find the category such as smooth or texture using the Mahalanobis
    distance measure.
Extraction of ResNet50 features
Apply PCA of ResNet50 features with 95% variance
fruit_berry_cnt ← 0;
foreach fruit_berry ∈ imade_dataset do
    if (fruit_berry == ((big andgreen and round and green_stem and smooth)or
    (big andgreen and oval and green_stem andsmooth) or (big andyellow and
    round and green_stem andsmooth) or (bigandyellow and oval and
    green_stem andsmooth)) then
        quality_grade ← A+;
    else if (fruit_berry == ((big andgreen and round and brown_stem
    andsmooth) or (big andgreen and oval and brown_stem andsmooth) or
    (big andyellow and round and brown_stem andsmooth) or (bigandyellow
    and oval and brown_stem andsmooth) or (small andgreen and round and
    green_stem andsmooth) then
        quality_grade ← A;
    else if (fruit_berry == ((small andgreen and oval and green_stem
    andsmooth) or (small andyellow and round and green_stem andsmooth)
    or (smalland yellow and oval and green_stem andsmooth) or (small
    andgreen and round and brown_stem andsmooth)) then
        quality_grade ← B;
    else if (fruit_berry == (((small andgreen and oval and brown_stem) or
    (small andyellow and round and brown_stem) or (smallandyellow and
    oval and brown_stem) or (small andgreen and oval and brown_stem))
    and(texture < 10%)) then
        quality_grade ← C;
    else
        quality_grade ← Rejected/D;
fruit_berry_cnt ←fruit_berry_cnt + 1
x ← quality_grade
end for
Stop
```

approach in terms of F1 score, overall accuracy, precision, recall and G-measure.

4.2.1. The performance comparison of transformation-based texture feature extraction techniques

Texture features are obtained by using transformed based techniques. Experimentation has been performed on DWT, Gabor filter and the combined approach of both of them. In DWT, most of the information is present in the LL sub-band but it does not provide any information regarding image singularity. The approximation coefficient of the next round, LL2 sub-band, is extracted for the next level decomposition. After the fourth level, it is perceived that information is getting lost, so for this study, the fourth level image decomposition of DWT is used as shown in Fig. 9. In this research, the DWT (db4) and Gabor filters which are explained in Section 3.5, are applied separately for the extraction of texture features to analyze the surface texture of fruit images. The well-liked statistical features are used efficiently to represent textures. Mahalanobis distance metric is utilized to classify textures for evaluating the fruit surface smoothness or roughness and the performance of this classification is evaluated through accuracy. The performance comparison of DWT, Gabor and hybrid DWT + Gabor filter-based

Fig. 8. Mango (top row) and grape (bottom row) sample hierarchical fruit representative images for quality categorization.

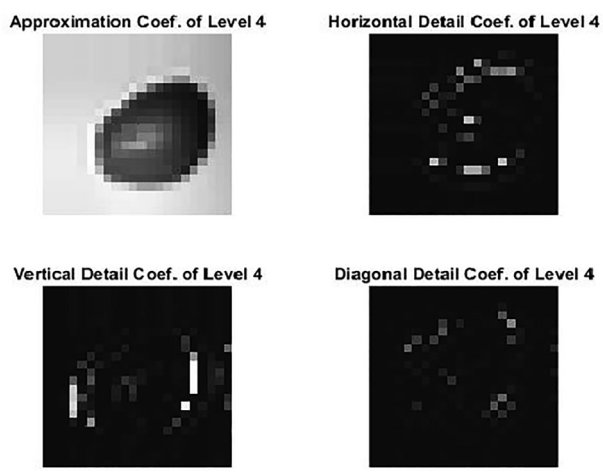


Fig. 9. Fourth level DWT decomposition.

texture feature extraction techniques for grape and mango fruits dataset are shown in Fig. 10.

From Fig. 10, it is observed that good results are achieved when DWT (db4) and Gabor filter (with frequency band 51–75%) are combined. Hence, the hybrid approach of the DWT + Gabor filter is used for texture feature extraction.

4.2.2. Fruits sorting method

In the experiment, the learning rate, epoch, batch size and dropout rate are set to 0.0001, 6, 32 and 0.5 respectively, and the cross-entropy loss function is selected. For the experimental purpose, the one-versus-all technique in SVM is used for the final fruit recognition. This technique develops a binary classifier

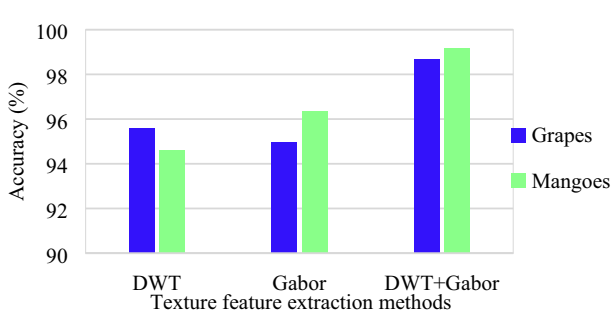


Fig. 10. Performance comparison of DWT, Gabor and DWT + Gabor filter-based texture feature extraction methods.

between any two classes of instances and decides which class the instance probably belongs to in each SVM in the training dataset.

The total features extracted using the ResNet50 model are 2048, whereas, for the Mango dataset, CP Softmax (co-occurrence pattern-based softmax) and CP_SVM (co-occurrence pattern-based SVM) models have a total of 345 features (SD + shape + size + two stem color features + four texture features + 336 deep features after applying PCA) respectively. For the Grape dataset, CP_SVM and CP_Softmax based models have a total of 140 input features (SD + shape + size + two stem color features + four texture features and 131 deep features after applying PCA) respectively. These features are fed to the SVM and softmax classifier for fruit annotation.

Tables 4 and 5 illustrate the F1 score values attained by ResNet50, CP_Softmax and CP_SVM. It is observed that features extracted using deep learning pretrained models are not sufficient to classify each grape type and each mango type. The combination of deep features with other mentioned features are more precise and reliable. The proposed CP based features are beneficial for two categories of grapes and 9 categories of mangoes. From Tables 6 and 7, it is depicted that the CP_SVM achieved a good F1 score as 0.973913 and 0.972477 for all grape categories of multi-classification. Whereas for mangoes, CP_SVM has comparable performance to CP_Softmax. CP_SVM and CP_Softmax have the best F1 score as 1 for Gulab_khus mango fruit. For Kesar and Totapuri, CP_SVM has the F1 score as 1, while for Baigan_phali, Hapus and Rajapuri fruits, it scored 0.99115, 0.968355 and 0.947368 respectively, which is better than CP_Softmax and ResNet50. For the Langra mango category, CP_Softmax scored 1, whereas, for Dashahari and Badami, it scored a good F1 score as 0.966443 and 0.931034 than other methods.

4.2.2.1. Statistical analysis. Kruskal-Wallis non-parametric ANOVA analysis is performed to detect inequalities between groups. It is an efficient rank dependent non-parametric method of one-way ANOVA used to find the significant difference between the observation of two or more groups at a time. Therefore, this testing has been used in the analysis of F1 score values among groups.

The hypothesis used for evaluation is considered as follows:

H0: there is no significant difference between the F1 score of groups.

H1: there is a significant difference between the F1 score of groups.

For the mango dataset, the statistic H value of mean rank equals to 14.70 as the computed H value exceeds the critical chi-square ($\chi 2$)) for 0.001 level, the probability of the NULL hypothesis being correct is considered to be low ($p \ll 0.001$). For the grape dataset, the computed H value = 6.7 is lower than $\chi 2$ for 0.05 level. Thus,

Table 4
Summary of F1 scores gained by each grape class.

Sr. No.	Class	ResNet50	CP_Softmax	CP_SVM
1	Milky_green	0.894737	0.954955	**0.973913**
2	Amber	0.890909	0.955752	**0.972477**

there are significant differences between the F1 score of groups. The critical chi-squares value with two degrees of freedom is quoted as $\chi 2(0.05(2) = 5.99)$, $\chi 2$ $(0.01(2) = 9.21)$ and $\chi 2$ $(0.001(2) = 13.82)$.

The overall accuracy of the Grape and Mango dataset for ResNet50, CP_Softmax, CP_SVM models are shown in Fig. 11. From this observation, it is perceived that the subcategories of grape and mango fruits are successfully classified with 97.3% and 97.2% accuracy using CP_SVM.

The evaluation measures in terms of precision, recall and G-measure of each grape and mango type using CP_Softmax and CP_SVM are shown in Tables 6 and 7 respectively.

The experimental results of quality category classification using CP are shown using the bar graph in Table 8. The extracted features from the fruit images are used to build the CP. These CP are further used by the classifier for hierarchical fruit annotation. Table 8 illustrate the co-occurrence pattern of quality category detection where image1 has size big, the color of the fruit is green, the stem is green, round shape and smooth is categorized as quality A+ grade of Milky-green grapes while image2 has small size, yellow color, green stem, round shape and smooth is categorized as quality B of Amber grapes. Similarly, the quality category of all fruits from another dataset (mango) is evaluated.

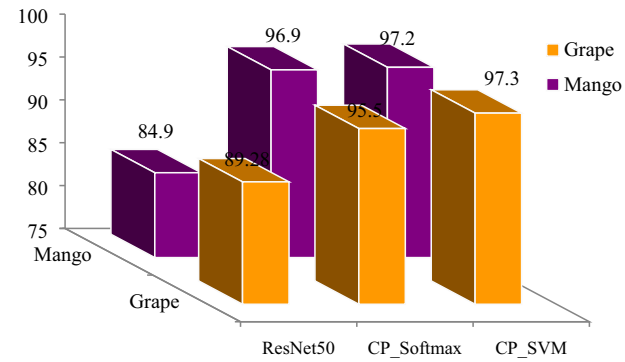


Fig. 11. Classification accuracy of ResNet50, CP_Softmax and CP_SVM for Grape and Mango datasets.

4.2.2.2. Comparison of results. From the various methods presented in the literature, only a few have received attention for quality grading and classification of grape and mango fruits, as shown in Table 9. Rajin et al. (2013) have classified the quality of uniformly colored fruits using the histogram and Canny edge detection method. Nandi et al. (2016) have graded mango fruit using a fuzzy-incremental learning algorithm based on maturity level and quality detection. Quality detection is accomplished through a Multi-attribute decision-making system on the basis of features such as shape, size and surface defect while Support Vector Regression (SVR) is utilized for finding maturity level prediction and achieved 87% accuracy. A noteworthy contribution to quality-based mango and guava fruit grade classification is done by

Table 5
Summary of F1 scores gained by each mango class.

Sr. No.	Class	ResNet50	CP_Softmax	CP_SVM
1	Badami	0.831169	**0.966443**	0.960526
2	Baingan_phali	0.918033	0.982143	**0.99115**
3	Dashehari	0.758621	**0.931034**	0.909091
4	Gulab_khas	0.939394	**1**	**1**
5	Hapus	0.835443	0.964706	**0.968355**
6	Kesar	0.835443	0.941176	**1**
7	Langra	0.935484	**1**	0.952381
8	Rajapuri	0.709677	0.945946	**0.947368**
9	Totapuri	0.837838	0.988235	**1**

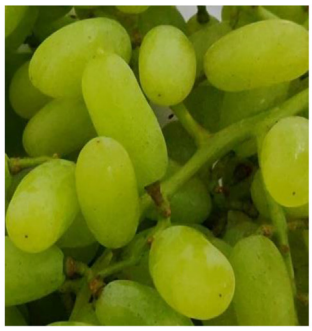
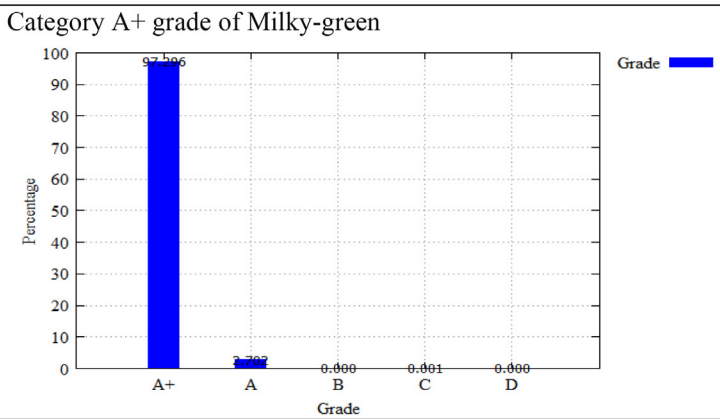
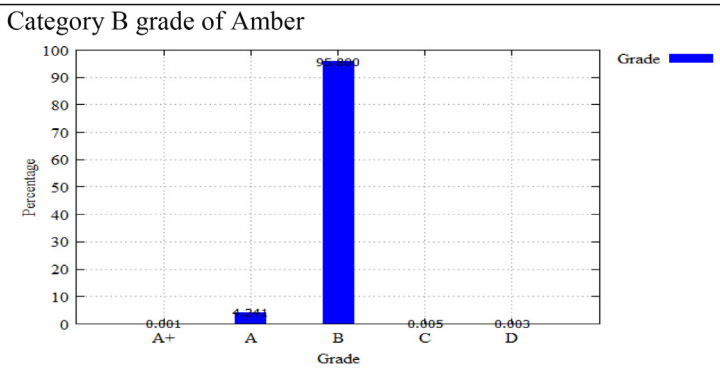
Table 6
Summary of precision, recall and G- measure for each grape class using ResNet50.

Sr. No.	Class	Precision		Recall		G-measure	
		CP_Softmax	CP_SVM	CP_Softmax	CP_SVM	CP_Softmax	CP_SVM
1	Amber	**0.963636**	0.949153	0.946429	**1**	0.954994	**0.974245**
2	Milky	0.947368	**1**	**0.964286**	0.946429	0.95579	**0.972846**

Table 7
Summary of precision, recall and G- measure for each mango class using ResNet50.

Sr. No.	Class	Precision		Recall		G-measure	
		CP_Softmax	CP_SVM	CP_Softmax	CP_SVM	CP_Softmax	CP_SVM
1	Badami	**0.96**	0.935897	0.972973	**0.986486**	**0.966465**	0.948132
2	Baingan_phali	**1**	**1**	0.964912	**0.982456**	0.982299	**0.995565**
3	Dashehari	0.964286	**1**	**0.9**	0.833333	0.931589	**0.953463**
4	Gulab_ khas	**1**	**1**	**1**	**1**	**1**	**1**
5	Hapus	0.953488	**0.97561**	**0.97619**	0.952381	0.964773	**0.969715**
6	Kesar	0.909091	**1**	0.97561	**1**	0.941763	**1**
7	Langra	**1**	0.9375	**1**	0.967742	**1**	0.944911
8	Rajapuri	**0.945946**	0.923077	0.945946	**0.972973**	**0.945946**	0.935144
9	Totapuri	**1**	**1**	0.976744	**1**	0.988304	**1**

Table 8
CP of quality category detection.

Sr. No.	Image	Quality Category classification
1		
2		

Khoje (2018). The extracted color and transform-based features are further optimized using a genetic algorithm and then classified by SVM and ANN classifiers which achieved 97.5% accuracy for guava fruits using SVM classifier and 95.6% accuracy for mango fruits using ANN classifier. Naik and Patel (2017) have contributed to grading the mango based on maturity level and size attributes, and then subsequently fuzzy rule-based classification is performed and achieved 89% accuracy. Qiaohua et al. (2017) have worked on

Table 9
Comparative analysis of grape and mango fruits sorting methods.

Author, Year	Fruit	No. of Samples	Feature Extraction	Classifier	Accuracy (%)
Rajin et al. (2013)	Grape	90	Histogram, Canny edge detection	PCA	96.6
Nandi et al. (2016)	Mango	200	Maturity level (actual-days-to-rot) and quality detection (shape, size and surface defect)	SVR, Multi-attribute decision making, Fuzzy-incremental learning algorithm	87
Khoje (2018)	Gauva Mango	435 345	Color and DWT based texture features are chosen through a genetic algorithm	SVM ANN SVM ANN	97.5 86.9 90.4 95.6
Naik and Patel (2017)	Mango	250	Maturity (color) and size (eccentricity, weight and area)	Fuzzy rule-based classification	89
Qiaohua et al. (2017)	Grape	A whole bunch of Red grapes	Color, edge and size	Distance measure	92.1
Gururaj and Viji (2019)	Mango	360	CNN As feature extractor	CNN-softmax	83
Kumari et al. (2019)	Mango	1200	Color, size, texture and weight	SVM	92
Raghavendra et al. (2020)	Mango	981	Color and texture	SVM	88
Proposed system	Grape Mango	400 1300	Color, texture, shape, size and deep features with PCA	Softmax, SVM Softmax, SVM	95.5 97.3 96.9 97.2

the gradation of a bunch of red globe grapes using color, edge and size features and attained 92.1% accuracy. CNN as feature extraction and classification is used by Gururaj and Viji (2019) for finding the maturity level which is used for quality gradation of mango fruit and achieved 83% accuracy. For quality testing and gradation of mangoes, Kumari et al. (2019), have used attributes such as color, size, texture and weight. Then SVM classifier is used for classification purposes and achieved 92% accuracy. Raghavendra et al. (2020) have used a hierarchical method for grading four classes of mangoes. The color features from L*a*b color space are extracted and classification is performed based on several classifiers such as LDA, Naïve Bayes, Threshold-Based, k-NN, PNN and SVM. The SVM classifier outperformed all other classifiers and attained 88% accuracy. The acquired accuracy of the proposed work is comparable with Khoje (2018) and Qiaohua et al. (2017). From Table 9, an improvement of 1.6% from 95.6% to 97.2% is observed for mango fruits, whereas for grape fruits there is an improvement of 5.2% from 92.1% to 97.3% which are stimulating and satisfactory. The obtained accuracy of the proposed system is better as compared to the existing available fruit sorting systems.

5. Conclusions

In this paper, a co-occurrence pattern for fruit quality and subcategory detection based on deep features, color, shape size and texture features are presented. The estimated size of the grape matches with the actual size of the grape verified by the vernier caliper. The distribution of RGB color space is observed using the color histogram, and then the color moment is computed to find color uniformity. The shape of the fruit image is identified as round or oval. The size of the fruit is evaluated based on the major axis and minor axis. The texture features are computed using a transform-based hybrid DWT + Gabor filter. The effective CNN features are extracted and their dimensions are reduced using PCA. Two datasets, one is binary (Grape dataset) and another multiclass (Mango dataset), are utilized for this experimentation. The main aim of this study is to investigate the performance of ResNet50, CP_Softmax and CP_SVM methods on both datasets. From the results, it is observed that the ResNet50 model provides less accuracy than CP_Softmax and CP_SVM. The CP_Softmax and CP_SVM performed well for the Mango dataset. CP_SVM model has given the F1 score for Milky_green and Amber grape types as 0.973913 and 0.972477 respectively, whereas for the Mango dataset CP_Softmax and CP_SVM model worked better than ResNet50 model. The overall quality categorization is improved after using the co-occurrence pattern, which makes it helpful for the in-line sorting of a fruit-based system. In the future, it is intended to apply the proposed method for other fruits also.

6. Financial and ethical disclosures

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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